DAY 1 · SESSION 2 · UNIVERSITY OF MINNESOTA

Long-Term Archival, Data Movement at Scale, and Hybrid Identity

Azure Storage, Data Protection, and Resilience at Scale

This afternoon, block by block

01 **Block 1 · 55 min**

Archival strategy and immutability foundations

02 **Block 2 · 45 min**

Data movement tooling and hybrid connectivity

03 **Break · 15 min**

Coffee and questions

04 **Block 3 · 50 min**

Hybrid identity and access governance, with demonstration

05 **Block 4 · 75 min**

Hands-on lab and debrief

Some university data must outlive every technology that stores it

- Some datasets carry retention expectations measured in centuries, and much of the data is irreplaceable
- The team needs an archival strategy that survives technology change, a plan to move petabytes, and an identity model that keeps control tight
- Builds directly on Session 1's tiers and lifecycle policies

By the end of this session you can

Design a long-term archival strategy using Archive tier, immutability, integrity verification, and format independence

Choose AzCopy, Storage Mover, Data Factory, or Data Box for a given volume, timeline, and network

Evaluate ExpressRoute versus VPN Gateway for sustained storage traffic

Configure identity-based access to Azure Files and Blob with RBAC and ABAC

Govern SAS and use managed identities so no workload depends on account keys

BLOCK 1 · 55 MIN

Archival strategy and immutability foundations

Retention, economics, integrity, and WORM policies that hold for decades.

Separate the retention decision from the technology decision

STABLE FOR DECADES

Retention decision

What must be kept, for how long, and why. Owned by data stewards and policy.

CHANGES MANY TIMES

Technology decision

Where and how it is stored today. Expected to change many times over the retention period.

Design so data can move between technologies without breaking its retention guarantees or its integrity record.

Archive tier economics over decades

1 **Offline, lowest cost**

Lowest storage cost, hours of retrieval latency, minimum 180 days.

2 **Capacity dominates**

Over decades, retrieval and rehydration are occasional, planned events.

3 **Budget the recurring**

Periodic rehydration tests and eventual format migrations are known costs.

4 **Bundle small files**

Many small blobs rehydrate slower; package them before archiving.

Integrity you can prove: manifests and checksum validation

- Create an integrity manifest at ingest: file path, size, and cryptographic hash for every object
- Store the manifest alongside the data and in a separate protected location
- Validate checksums periodically on a sample and fully during any migration
- Record every validation result as audit evidence

Rehydration testing and format migration planning

01

Schedule tests

Run rehydration on a regular cadence, not only when data is requested.

02

Measure time

Compare real retrieval time with the up to 15 hours of standard priority.

03

Validate

Check restored data against the manifest.

04

Plan migrations

Track formats at risk of obsolescence; move to open, documented formats.

Immutable Blob storage: write once, read many

WORM state: data can be created and read but not modified or deleted for a user-specified interval.

Time-based retention

After the period expires, objects can be deleted but not overwritten.

Legal hold

Data stays immutable until the hold is explicitly cleared.

Combined

Both policies can apply at the same time and level.

Container-level or version-level WORM

CONTAINER-LEVEL

Policy on the container only

- Retention between 1 and 146,000 days (400 years)
- Supported on GPv2, premium block blob, and hierarchical namespace accounts; immutable blobs on HNS accounts cannot be renamed or moved
- Up to 10,000 containers with a policy per storage account

VERSION-LEVEL

Account, container, or blob version

- Requires blob versioning
- Supported on GPv2 and premium block blob accounts
- Not supported on hierarchical namespace accounts, because versioning is not available there

Neither policy type is supported on accounts with NFS 3.0 or SFTP enabled, and immutable storage cannot be combined with last access time tracking or point-in-time restore.

Locked versus unlocked: the decision you make once

UNLOCKED

For testing

The retention interval can be changed and the policy deleted. Blobs are protected from overwrite and delete while the policy is active, but the storage account itself can still be deleted.

LOCKED

No deletion. No shortening.

The policy cannot be deleted; a container-level policy can be extended up to five times, never shortened. The storage account cannot be deleted while any container carries a locked policy.

Up to 35 seconds to take effect. Writes and deletes may succeed during activation, and a visible policy on the container is not proof that it has taken effect.

Microsoft's guidance: test unlocked, then lock within about 24 hours.

Locking for centuries means storage charges for centuries.

Validate scope and interval before locking.

BLOCK 1 CHECKPOINT

Three archival questions

01 Who owns the retention decision for each dataset?

02 How will you prove integrity in 20 years?

03 Which datasets justify a locked policy, and which a legal hold?

BLOCK 2 · 45 MIN

Data movement tooling and hybrid connectivity

Tools, links, and costs for moving petabytes.

Choose the tool from volume, timeline, and network

Microsoft's selection guidance weighs data size, available bandwidth, and whether transfer is one-time or recurring.

Online

AzCopy, Azure Storage Mover, Azure Data Factory.

Offline

The Data Box family, when the network cannot move the volume in time.

File Sync

Azure File Sync can migrate to Azure Files with no downtime.

AzCopy: the fast, scriptable workhorse

Command line

Copies to and from Blob and Azure Files, and between accounts.

Protocol paths

Copies between local or mounted file systems, Blob, and Azure Files, including SMB and NFS shares.

Throughput

Tune concurrency and buffers; use sync for incremental runs.

Authentication

Use Microsoft Entra ID rather than keys where possible.

Azure Storage Mover: managed migration at scale

Fully managed

Terabyte to petabyte scale over the network.

Two parts

Cloud service plus an on-premises migration agent VM.

One dashboard

One resource orchestrates multiple agents.

Repeatable

Lift-and-shift and periodic migrations of NFS and SMB shares.

Azure Data Factory: orchestrated and recurring transfers

Orchestrate

Pipelines schedule and orchestrate copies across many sources and sinks.

Reach on premises

Self-hosted integration runtime reaches on-premises data stores.

Transform

Fits recurring feeds, transformations, and multi-step workflows.

Monitor and retry

Monitoring and retry built into pipeline runs.

Offline transfer with the Data Box family

Microsoft ships a ruggedized device; you copy data locally and ship it back for upload into your account.

- **Data Box Disk** · SSDs for smaller volumes
- **Data Box 120:** next-generation appliance, 120 TB usable, network interfaces up to 100 Gbps
- **Data Box 525:** next-generation appliance, 525 TB usable; order several for petabyte moves
- **Retired:** Data Box Heavy is no longer orderable, and the original 80 TB Data Box is being retired region by region as the next-generation devices roll out; existing orders remain supported.

Check current device capacities and regional availability in Microsoft Learn before planning.

Matching tool to scenario

Scenario	Recommended tool
One-off copy under 1 TB, scripted	AzCopy
NFS or SMB share migration, TB to PB, over network	Azure Storage Mover
Recurring orchestrated feeds with transforms	Azure Data Factory
Windows file servers to Azure Files with ongoing cache	Azure File Sync
Petabytes with limited bandwidth or tight deadline	Data Box 525 (several orders) or Data Box 120

ExpressRoute or VPN Gateway for sustained storage traffic

Compare	ExpressRoute	VPN Gateway
Connection	Private connection through a connectivity provider	Encrypted tunnels over the internet
Performance	Predictable latency; circuit bandwidths sized for sustained transfer	Aggregate throughput limited by gateway SKU
Peering	Microsoft peering reaches public Microsoft service endpoints; private peering reaches virtual networks and private endpoints	-

Private endpoints work over either link, keeping storage off the public internet.

Know where egress charges apply

Direction matters

Data into Azure is generally free; data out of Azure regions is billed.

ExpressRoute plans

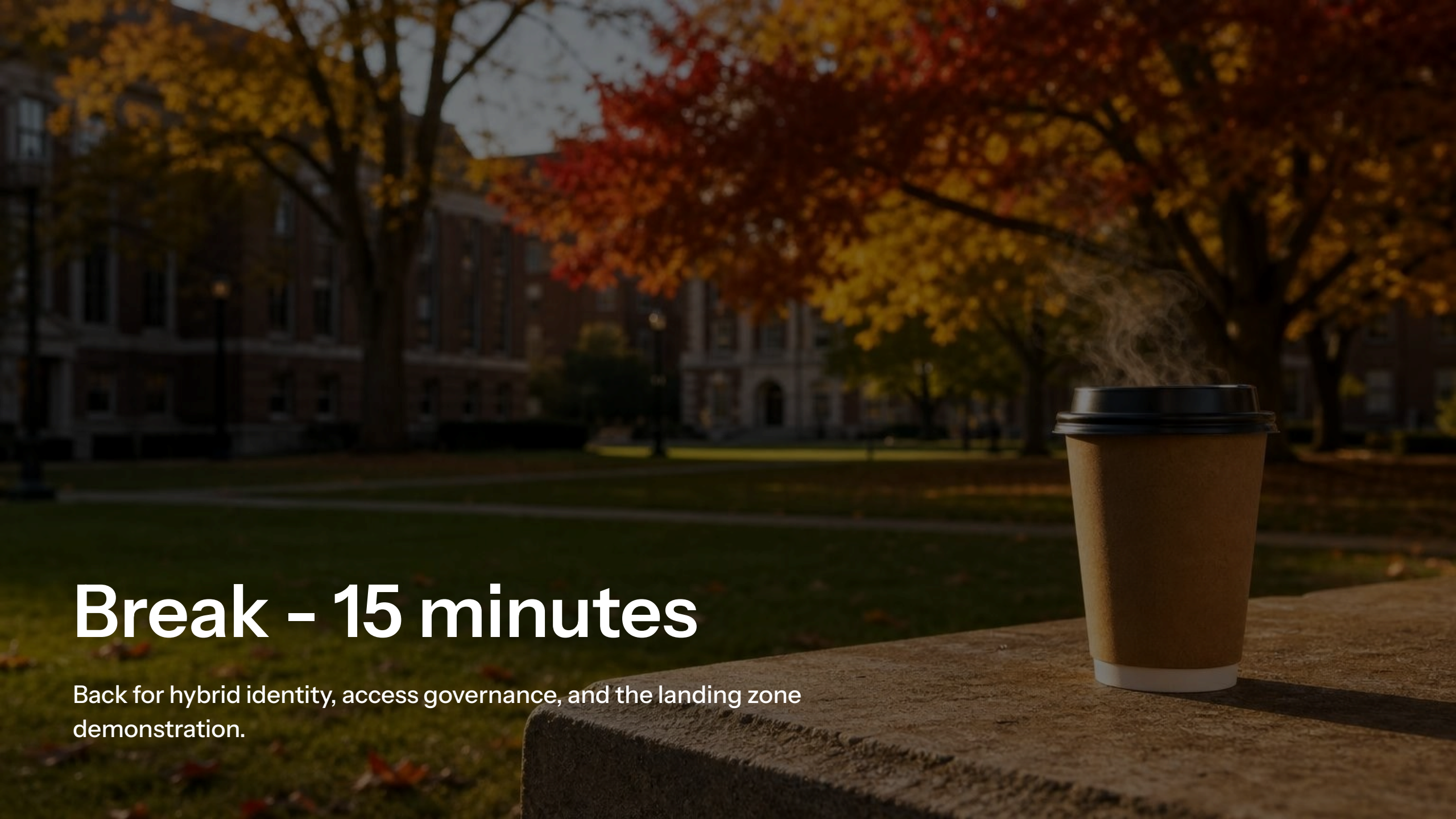
Metered and unlimited data plans price outbound data differently.

Cross-region

Replication and reads also carry transfer charges.

Model first

Before designs that routinely pull data back on premises.



Break - 15 minutes

Back for hybrid identity, access governance, and the landing zone demonstration.

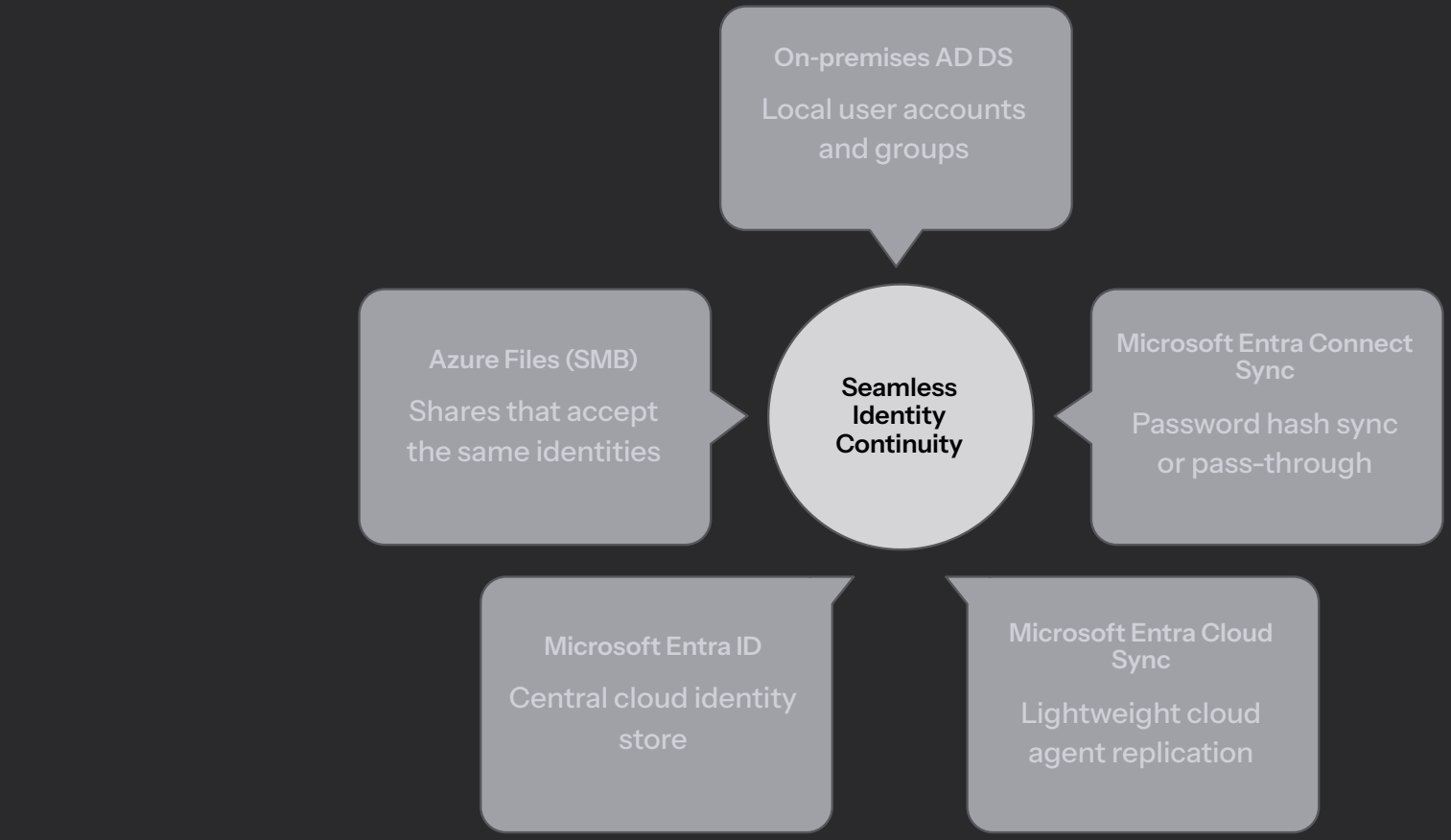
BLOCK 3 · 50 MIN

Hybrid identity and access governance

With demonstration: one identity, precise access, no keys to leak.

One identity, on premises and in the cloud

Identity-based authentication is preferred over storage account keys for security.



Azure Files: three identity sources, one per account

On-premises AD DS

Entra Domain Services

Entra Kerberos

For hybrid and cloud-only identities.

- Only one identity source can be enabled per storage account
- Kerberos with AES-256 encryption is recommended
- Applies to SMB shares, not NFS shares

Share-level and file-level permissions work together

SHARE LEVEL · AZURE RBAC

A default permission for all authenticated users, or granular roles for synced users and groups.

Groups created only in Entra ID work only if they contain synced users.

DIRECTORY AND FILE · WINDOWS ACLS

Set as on today's file servers.

With Entra Kerberos for hybrid identities, setting ACLs needs line of sight to a domain controller.

Blob access with Microsoft Entra ID and RBAC

Built-in data roles

Storage Blob Data Reader, Contributor, and Owner.

Scope

Assign at account, container, or resource group scope, ideally to groups.

Data plane is separate

Distinct from management-plane roles such as Contributor.

HNS refinement

On HNS accounts, POSIX ACLs refine access below RBAC.

Attribute-based access control: precise access without ACL sprawl

ABAC adds conditions to role assignments based on attributes such as blob index tags or container names.

EXAMPLE CONDITION

Grant Storage Blob Data Reader only where the blob's research unit tag matches the user's unit

- One condition replaces many per-folder ACLs
- Tags set in Session 1 for lifecycle now drive access too

Managed identities: automation with no secrets to leak

1

Assign an identity

System-assigned or user-assigned, for VMs, Functions, and Data Factory.

2

Grant a data role

Tokens are issued and rotated by the platform.

3

Remove secrets

No account keys or connection strings in scripts and pipelines.

Shared access signatures: govern them or disable them

RECOMMENDED

User delegation SAS

Signed with Entra ID credentials.

KEY-SIGNED

Service and account SAS

Signed with the account key.

- Use stored access policies to revoke service SAS centrally; keep expiry short
- Rotate keys regularly, then disable shared key access at the account level to block key-signed SAS entirely

DEMONSTRATION

The century-proof landing zone

1	2	3	4
Deploy an archival account with a locked time-based immutability policy	Upload a synthetic satellite dataset with its integrity manifest	Attempt to overwrite and delete it	Grant a synchronized user access through an ABAC condition scoped to a research unit tag

Watch the platform itself refuse the delete, and one condition grant precise access without touching an ACL.

BLOCK 4 · 75 MIN

Hands-on lab and debrief

Build it yourself, then compare notes.

Lab: build an archival landing zone with identity-based access

01	02	03	04	05
Deploy a standard GPv2 storage account with a flat namespace and zone redundancy (the archive pattern from Session 1)	Create a container with an unlocked time-based retention policy	Enable blob versioning and add a lifecycle rule to Archive based on creation or modification time	Assign Storage Blob Data Reader with an ABAC condition on a research unit index tag and validate from a user account	Submit a rehydration request to experience Archive latency

Lab notes: start long operations early

- Start the rehydration request first and revisit it at debrief
- Keep the retention policy unlocked so lab resources can be cleaned up; an unlocked policy still blocks blob deletes, so remove the policy, then empty the container, then delete the account
- The lab uses a flat namespace on purpose: blob versioning and version-level WORM are not supported on hierarchical namespace accounts, and blob index tags are preview there
- Do not enable last access time tracking on this account; it is incompatible with immutable storage, which is why the lifecycle rule uses creation or modification time
- Use the validation checkpoints in the lab guide after each step

Debrief questions

1. Where did the platform enforce a guarantee you rely on process for today?
2. How long did rehydration actually take, and what does that mean for researcher expectations?
3. Which ABAC condition would replace the most ACLs in your environment?

Key takeaways and what comes next

Archival is a strategy: retention, integrity, immutability, and format migration together

Pick movement tools by volume, timeline, and network; model egress

Identity, ABAC, and managed identities replace keys and ACL sprawl

Materials: demo script, lab guide, data movement decision tree, sample integrity manifest and validation script

Capstone input: Century-proof archival landing zone with integrity validation

Next, Day 2: data protection in both directions